NORMALITY ASCENDS ALONG WEAKLY ETALE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19150-19154 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19156-19162, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Seventh weakly etale-series card; the comparison lemma it cites is that series' local-rings pillar (already in the published chapter body, tag-gated clean).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma henselization-normal (strict-henselian criterion for normality); chapter lemma weakly-etale-strictly-henselian-local-rings (weakly etale maps are iso on strict henselizations).

Source reference. Original lemma 094Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19098–19106. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06DI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19162–19162. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. If Y is a normal scheme and f weakly étale, then X is a normal scheme.*

Proof. By More on Algebra, Lemma 06DI a scheme S is normal if and only if for all $s \in S$ the strict henselization of $\mathcal{O}_{S,s}$ is a normal domain. Hence the lemma follows from Lemma 094Z. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.